

Surface-Code Syndrome Utilization for Fault Identification

Can repeated stabilizer syndromes reveal *where* a device is faulty—not just *that* it is?

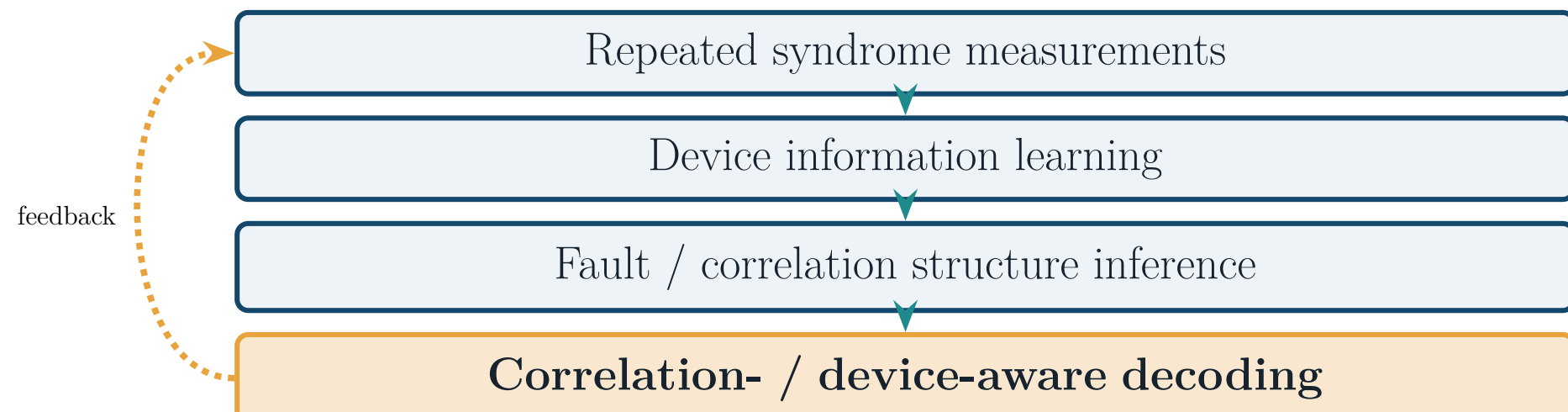
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1. Motivation

Surface codes already measure syndromes every round—thousands of stabilizer bits are produced purely to detect and decode errors, then discarded.

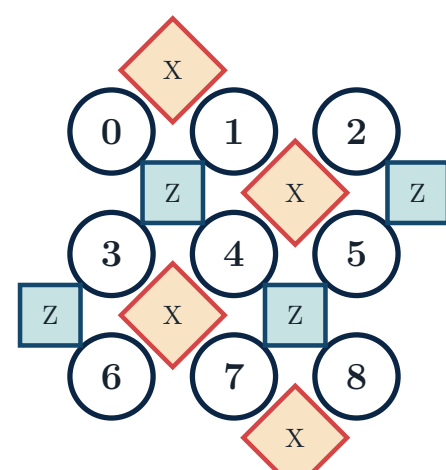
“This syndrome stream is a free probe of the device. Can we read its own fault structure out of data we already collect for QEC?”

Long-term vision: a **closed loop** that recycles QEC data into device knowledge, then feeds it back into better decoding.



2. Case Study & Problem

Setting. Distance-3 rotated surface code (17 qubits: 9 data + 8 ancilla); each stabilizer-extraction round issues **24 directed CNOTs**. One CNOT is the *dominant* fault source (higher fault probability than the rest).



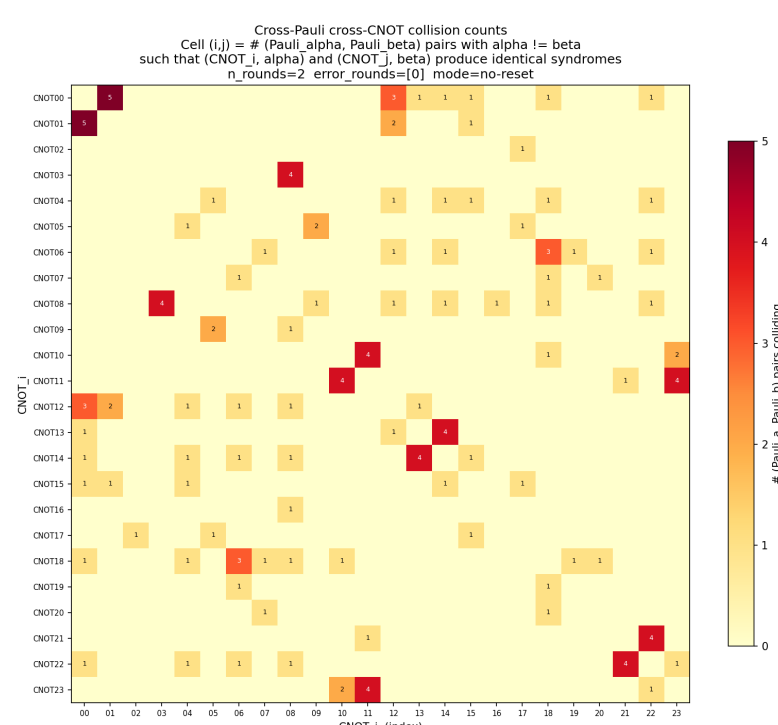
Rotated $d=3$ layout: 9 data qubits (circles), 8 Z-/X-stabilizer ancillas (squares / diamonds).

The question

From a temporal syndrome / detection-event sequence $S = (s_1, \dots, s_T)$ *alone*, identify the **dominant faulty CNOT** among the 24. **Nuisance factors** (unobserved): which rounds the fault fires, and its Pauli type ($XX, \dots, ZZ$).

3. Single-Shot Ambiguity

Is one measurement enough? **No**. When the Pauli type is unknown, distinct (CNOT, Pauli) faults can produce *identical* syndromes. Enumerating all 216 single-round cases:



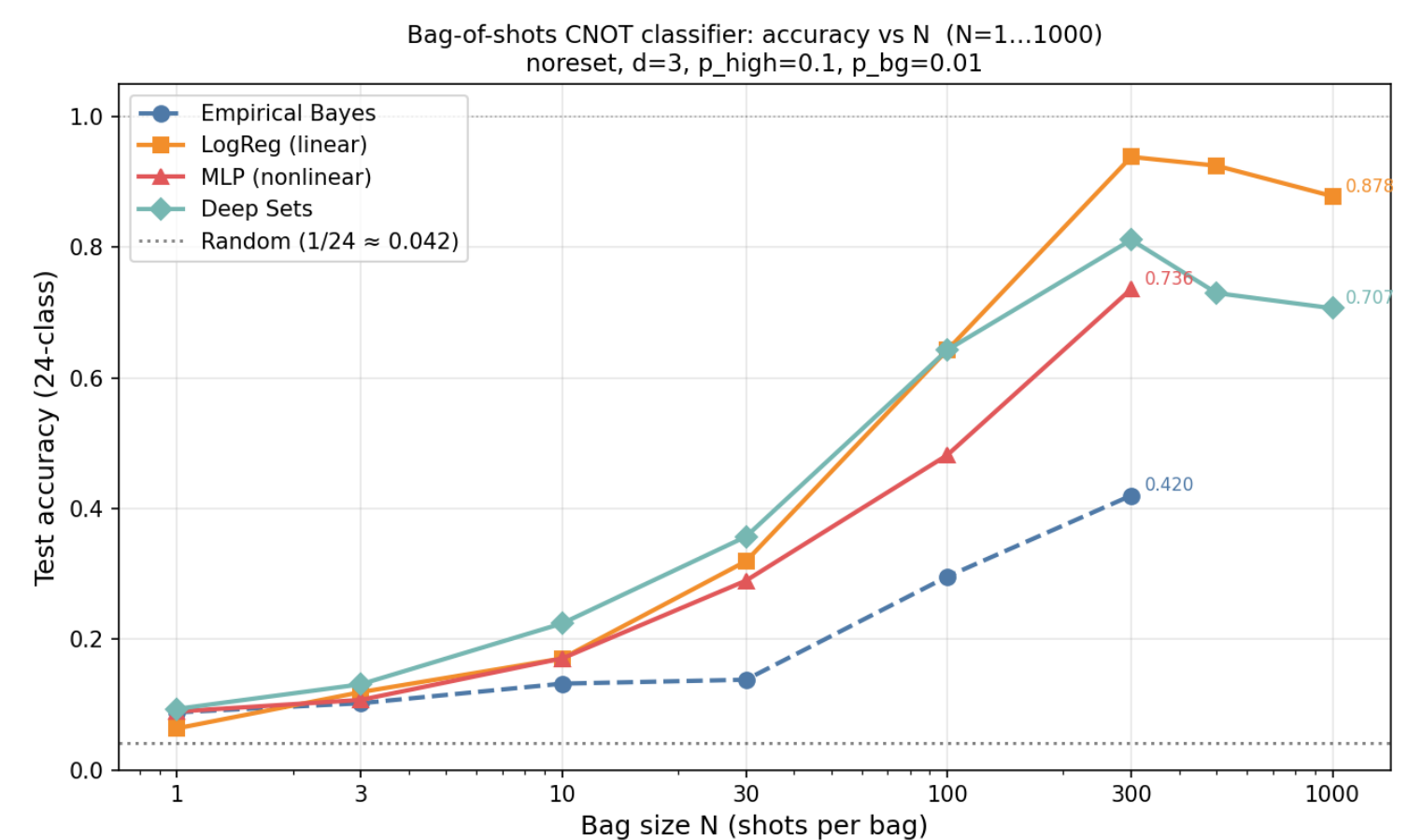
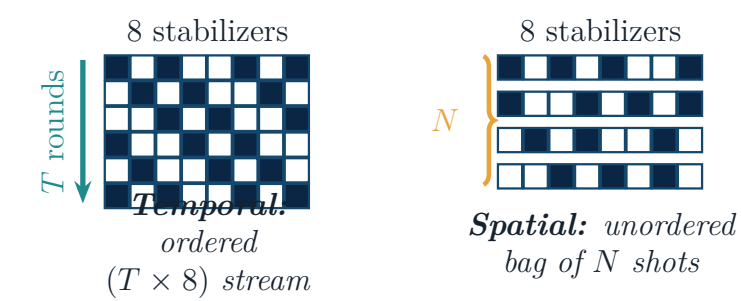
Collision map: cell (i, j) counts Pauli pairs for which faults at $CNOT_i$ and $CNOT_j$ are indistinguishable.

Hard limit

216 cases collapse to 123 signatures with **72** cross-CNOT collisions; a single round with known Pauli identifies only 8/24. **Single-shot 24-class identification is information-theoretically impossible**—invariant under round count and reset mode.

4. More Data Breaks the Ambiguity

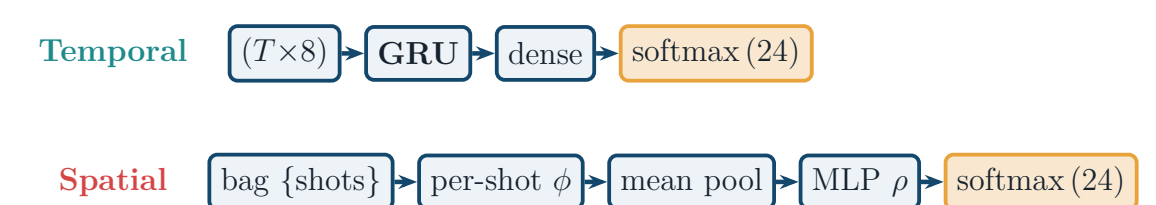
The way out is **aggregation**. The same fault is probed either *temporally* (a long syndrome stream) or *spatially* (many repeated single-round shots):



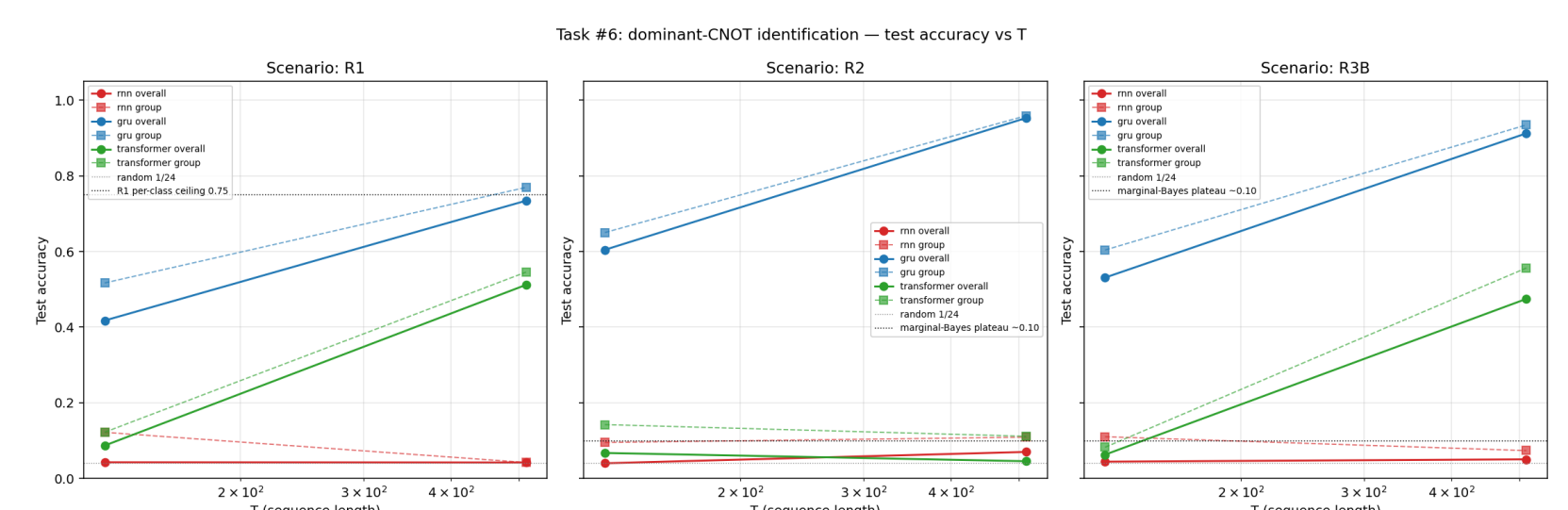
Bag-of-shots CNOT accuracy vs bag size N . At $N=1$ (single shot) accuracy is near random ($1/24$); by $N=300$ it reaches ≈ 0.9 —the ambiguity is broken by data volume, not by a better single shot.

5. Learned Decoders: Models & Performance

Each regime has a matched model class. Best performers so far (**preliminary**, naive $d=3$):



Deep Sets (spatial) is permutation-invariant over the bag; a linear LogReg on pooled features is a strong baseline.



Temporal accuracy vs sequence length T (three fault-model scenarios). The **GRU** (blue) climbs with T to $\approx 95\%$ (R2), breaking the rule-based single-round ceiling (0.75); RNN / transformer lag.

Headline (preliminary)

Temporal GRU $\approx 95\%$; spatial LogReg $\approx 94\%$ at $N=300$. Reported as in-progress, not final performance claims.

6. Status & Outlook

Done. Exact structural map of what syndromes can / cannot reveal about fault location at $d=3$.

Ongoing. Stim-based *realistic* circuit (parallel schedule, general d); *probabilistic* dominant-fault model.

Next. Close the loop—feed inferred fault / correlation structure into **device-aware decoding**.